

## Exponential functions



1 Consider the function  $y = 5 \times 4^x$ . Find the value of  $y$  when:

a  $x = 0$

b  $x = 2$

c  $x = -1$

d  $x = 3$

2 Consider the function  $f(x) = \left(\frac{4}{3}\right)^x$ .

a Evaluate  $f(0)$ .

b Evaluate  $f(-2)$ .

3 Find the value of  $5000(1.08^x)$  for  $x = 30$  to two decimal places.

4 Find the missing coordinate in each ordered pair that is a solution to  $y = 5^x$ :

a  $(3, 1)$

b  $(1, 5)$

c  $(-1, 1)$

d  $\left(1, \frac{1}{25}\right)$

5 Find the missing coordinate in each ordered pair that is a solution to  $y = -2^x$ :

a  $(5, 1)$

b  $\left(1, -\frac{1}{8}\right)$

c  $(-1, 1)$

d  $(1, -16)$

6 If the point  $(3, m)$  lies on the curve  $y = 2^x$ , solve for  $m$ .

7 The points  $(3, n)$ ,  $(k, 16)$  and  $\left(m, \frac{1}{4}\right)$  all lie on the curve with equation  $y = 2^x$ .

a Solve for  $n$ .

b Solve for  $k$ .

c Solve for  $m$ .

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## Characteristics of exponential functions

8 Do either of the functions  $y = 9^x$  or  $y = 9^{-x}$  have  $x$ -intercepts? Explain your answer.

9 For what values of  $x$  will  $6^x$  have a negative value?

10 Determine the  $y$ -intercept of all exponential functions of the form:

a  $y = a^x$

b  $y = a^{-x}$

c  $y = -a^x$

d  $y = -a^{-x}$



**11** Consider an exponential curve with equation of the form  $y = a^x$ , where  $a > 0$ .

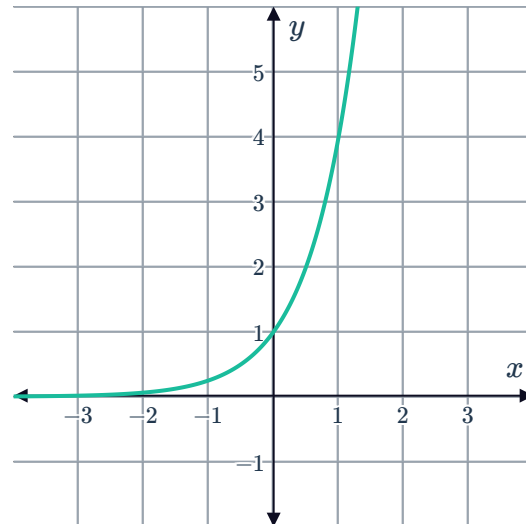
- a** What is the  $y$ -value of the  $y$ -intercept of all exponential curves of this form?
- b** What is the  $y$ -value of the  $y$ -intercept of  $y = (3.5)^x$ ?

**12** Consider the expression  $3^x$ .

- a** Evaluate the expression when  $x = -4$ .
- b** Evaluate the expression when  $x = 0$ .
- c** Evaluate the expression when  $x = 4$ .
- d** What happens to the value of  $3^x$  as  $x$  gets larger?
- e** What happens to the value of  $3^x$  as  $x$  gets smaller?

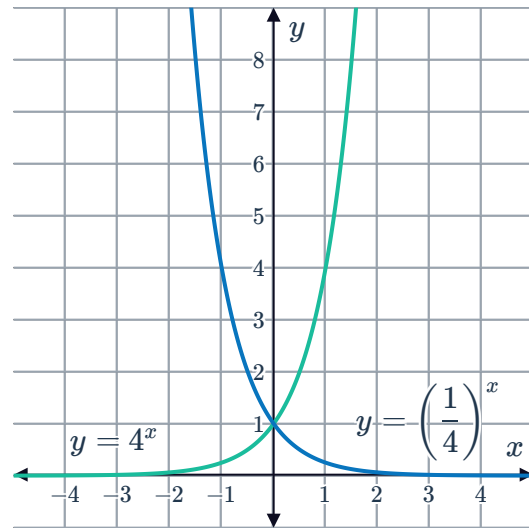
**13** Consider the graph of the function  $y = 4^x$ :

- a** Is each  $y$ -value of the function positive or negative?
- b** State the value of  $y$  the graph approaches but does not reach.
- c** State the equation and name of the horizontal line, which  $y = 4^x$  gets closer and closer to but never intersects.



**14** Given the function  $y = -10^x$ , what is the largest possible function value? Explain your answer.

- 15 Consider the graphs of the functions  $y = 4^x$  and  $y = \left(\frac{1}{4}\right)^x$ .



- Which function is an increasing function?
- How would you describe the rate of increase of  $y = 4^x$ ?

- 16 Of the two functions  $y = 4^x$  and  $y = 5^x$ , which increases more rapidly for  $x > 0$ ?

- 17 Consider the two functions  $y = 4^{-x}$  and  $y = 5^{-x}$ . Which function decreases more rapidly for  $x > 0$ ?

- 18 Consider the function  $y = 3^x$ .

- Complete the table of values:

|     |    |    |    |    |    |   |   |   |   |   |   |    |
|-----|----|----|----|----|----|---|---|---|---|---|---|----|
| $x$ | -5 | -4 | -3 | -2 | -1 | 0 | 1 | 2 | 3 | 4 | 5 | 10 |
| $y$ |    |    |    |    |    |   |   |   |   |   |   |    |

- Describe the behavior of the function as  $x$  increases.
- State the domain of the function.
- State the range of the function.

19 Consider the function  $y = 2^{-x}$ .



a Complete the table of values:

|     |    |    |    |    |    |   |   |   |   |   |   |    |
|-----|----|----|----|----|----|---|---|---|---|---|---|----|
| $x$ | -5 | -4 | -3 | -2 | -1 | 0 | 1 | 2 | 3 | 4 | 5 | 10 |
| $y$ |    |    |    |    |    |   |   |   |   |   |   |    |

b Describe the behavior of the function as  $x$  increases.

c Find the  $y$ -value of the  $y$ -intercept of the function.

d State the domain of the function.

e State the range of the function.

20 Consider the function  $y = -(2^x)$ .

a Complete the table of values:

|     |    |    |    |    |    |   |   |   |   |   |   |
|-----|----|----|----|----|----|---|---|---|---|---|---|
| $x$ | -5 | -4 | -3 | -2 | -1 | 0 | 1 | 2 | 3 | 4 | 5 |
| $y$ |    |    |    |    |    |   |   |   |   |   |   |

b Can the value of  $y$  ever be zero or positive? Explain your answer.

c Is  $y = -(2^x)$  an increasing or decreasing function?

d Describe the behavior of the function as  $x$  increases.

e State the domain of the function.

f State the range of the function.

21 Consider the graph of the function  $y = 2^x$ :

a State the  $y$ -intercept of this graph.

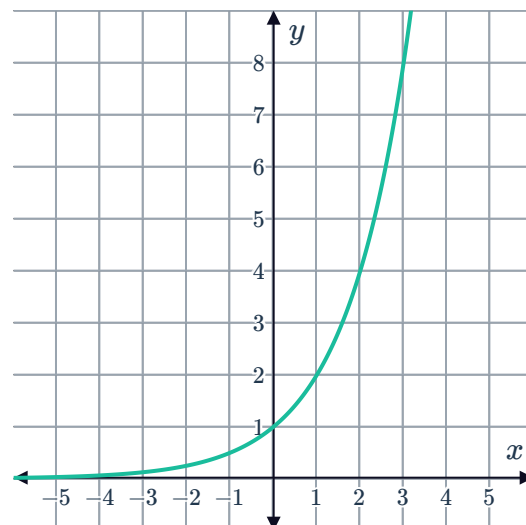
b Does the graph have an  $x$ -intercept?

c State the domain of the function.

d State the range of the function.

e Find the value of  $y$  when  $x = 7$ .

f Find the value of  $x$  when  $y = 256$ .



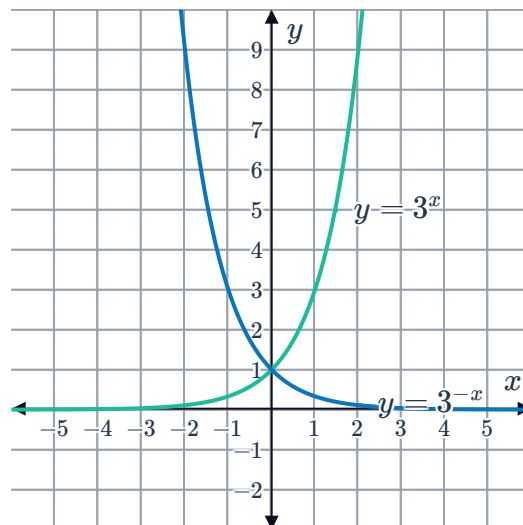
**22** Consider the functions  $y = 2^{-x}$ ,  $y = 3^{-x}$  and  $y = 5^{-x}$ .



- a** State a characteristic that all the graphs of the functions have in common.
- b** What is the  $y$ -value of the shared  $y$ -intercept?
- c** Describe the nature of these functions for large values of  $x$ .

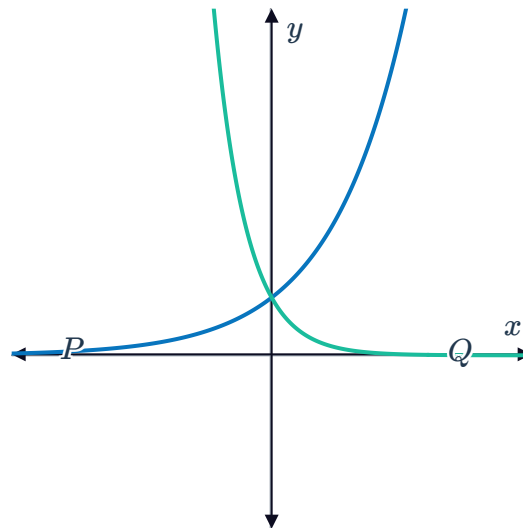
**23** Consider the graph of the following functions  $y = 3^x$  and  $y = 3^{-x}$ :

- a** State the coordinates of the point of intersection of the two curves.
- b** Describe the behavior of both these functions for large values of  $x$ .



**24** Consider the given graphs of the two exponential functions  $P$  and  $Q$ :  
State whether the following pairs of equations could be the equations of the graphs  $P$  and  $Q$ :

- a**  $P : y = 2^x$   
 $Q : y = 2^{-x}$
- b**  $P : y = (3.5)^x$   
 $Q : y = 6^{-x}$
- c**  $P : y = 2^x$   
 $Q : y = 5^{-x}$
- d**  $P : y = 5^x$   
 $Q : y = 2^{-x}$

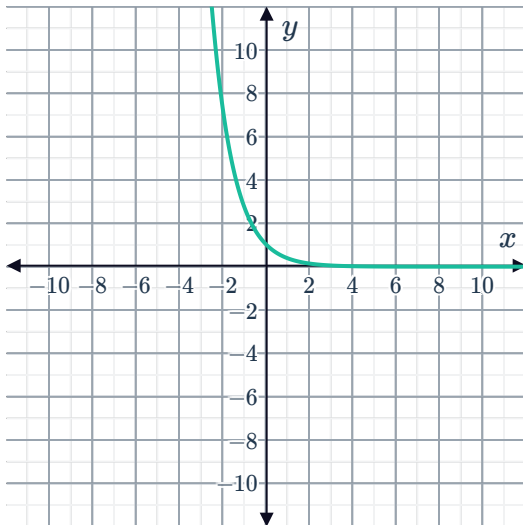




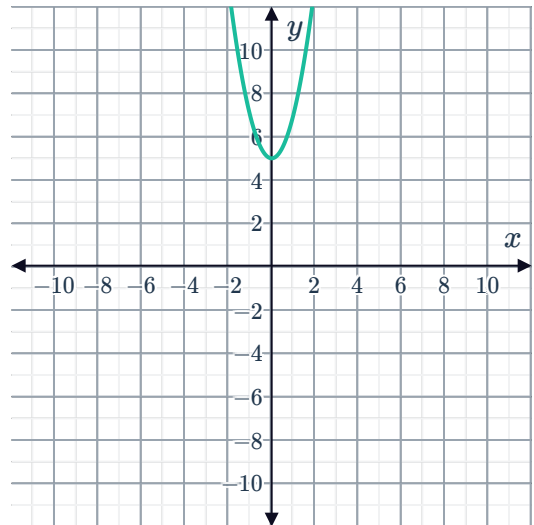
## Graphs of exponential functions

**25** Determine whether the following graphs show an exponential function:

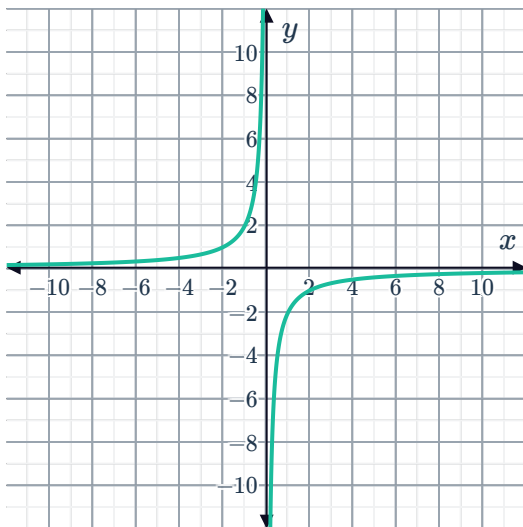
**a**



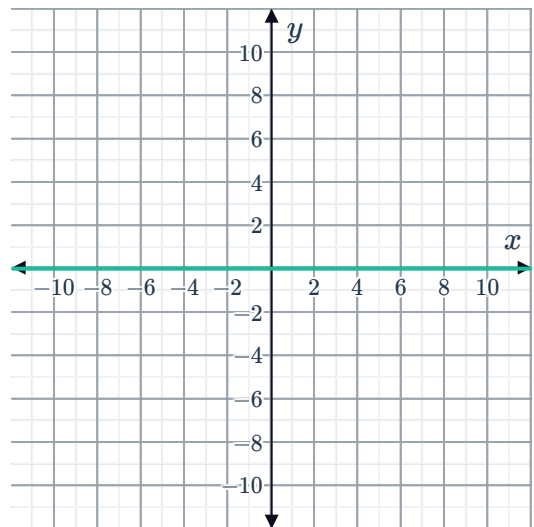
**b**



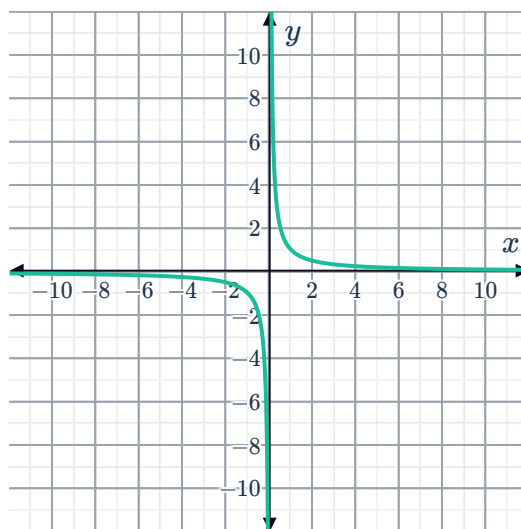
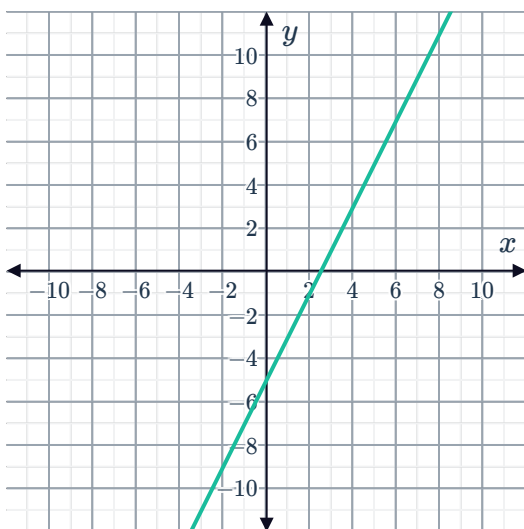
**c**



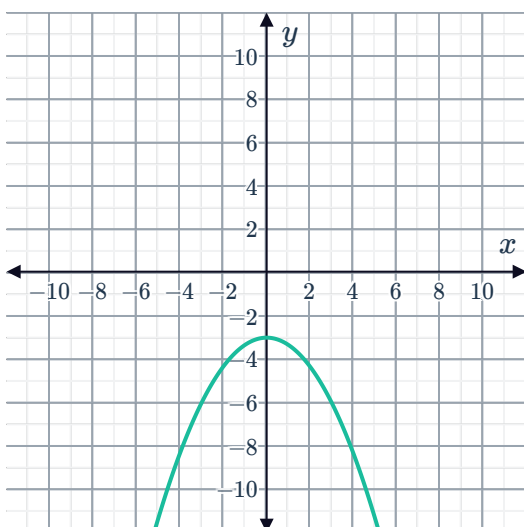
**d**



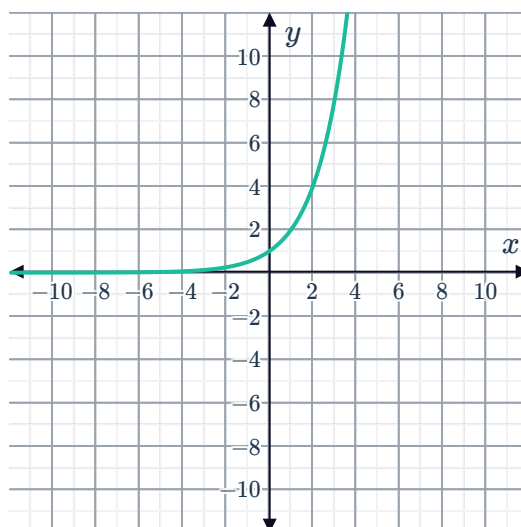
e



g



h



**26** Consider the equation  $y = 3^x$ .

- Complete the table of solutions for the given equation.
- Plot the points  $C$ ,  $D$ , and  $E$  on a coordinate plane.
- Sketch the curve that results from the entire set of solutions for the equation being graphed.

| Point | Coordinates          |
|-------|----------------------|
| $A$   | $(-3, \frac{1}{27})$ |
| $B$   | $(-2, \frac{1}{9})$  |
| $C$   | $(-1, \frac{1}{3})$  |
| $D$   | $(0, 1)$             |
| $E$   | $(1, 3)$             |
| $F$   | $(2, 9)$             |
| $G$   | $(3, 27)$            |

27 Consider the equation  $y = 2^x$ .



a Complete the table of values:

|     |    |    |    |   |   |   |   |
|-----|----|----|----|---|---|---|---|
| $x$ | -3 | -2 | -1 | 0 | 1 | 2 | 3 |
| $y$ |    |    |    |   |   |   |   |

- b Sketch the graph of the function.
- c What happens to the  $y$ -values as  $x$  increases?
- d What happens to the  $y$ -values as  $x$  decreases?
- e Does the curve cross the  $x$ -axis?
- f At what value of  $y$  does the graph cross the  $y$ -axis?

28 Consider the equation  $y = 2^{-x}$ .

a Complete the table of values:

|     |    |    |    |   |   |   |   |
|-----|----|----|----|---|---|---|---|
| $x$ | -3 | -2 | -1 | 0 | 1 | 2 | 3 |
| $y$ |    |    |    |   |   |   |   |

- b Sketch the graph of the function.
- c What happens to the  $y$ -values as  $x$  increases?
- d What happens to the  $y$ -values as  $x$  decreases?
- e Does the curve cross the  $x$ -axis?
- f At what value of  $y$  does the graph cross the  $y$ -axis?

29 For each of the following equations:

- i Find the  $y$ -intercept of the curve.
- ii Complete the table of values:

|     |    |    |    |   |   |   |   |
|-----|----|----|----|---|---|---|---|
| $x$ | -3 | -2 | -1 | 0 | 1 | 2 | 3 |
| $y$ |    |    |    |   |   |   |   |

- iii Find the horizontal asymptote of the curve.
- iv Sketch the graph of the function.

a  $y = 6^x$

b  $y = 3^x$

c  $y = 10^x$

d  $y = 3^{-x}$



**30** Consider the function  $y = 6^x$ .

- a** Can the value of  $y$  ever be zero or negative? Explain your answer.
- b** As the values of  $x$  get larger and larger, what value does  $y$  approach?
- c** As the values of  $x$  get smaller and smaller, what value does  $y$  approach?
- d** Find the  $y$ -value of the  $y$ -intercept of the curve.
- e** How many  $x$ -intercepts does the curve have?
- f** Sketch the graph of the function.

**31** Consider the function  $y = 8^{-x}$ .

- a** Can the value of  $y$  ever be zero or negative? Explain your answer.
- b** As the value of  $x$  gets larger and larger, what value does  $y$  approach?
- c** As the value of  $x$  gets smaller and smaller, what value does  $y$  approach?
- d** Find the  $y$ -value of the  $y$ -intercept of the curve.
- e** How many  $x$ -intercepts does the curve have?
- f** Sketch the graph of the function.

**32** Consider the function  $y = -3^x$ .

- a** Find the  $y$ -intercept of the curve  $y = -3^x$ .
- b** Find the horizontal asymptote of the curve  $y = -3^x$ .
- c** Now, sketch the graph of the function.
- d** Is the function  $y = -3^x$ , an increasing or decreasing function?

**33** Consider the function  $y = -6^x$ .

- a** Find the  $y$ -intercept of the curve  $y = -6^x$ .
- b** Find the horizontal asymptote of the curve  $y = -6^x$ .
- c** Now, sketch the graph of the function.
- d** Is the function  $y = -6^x$ , an increasing or decreasing function?

34 Consider the function  $y = -3^{-x}$ .



- a Find the  $y$ -intercept of the curve  $y = -3^{-x}$ .
- b Find the horizontal asymptote of the curve  $y = -3^{-x}$ .
- c Now, sketch the graph of the function.
- d Is the function  $y = -3^{-x}$ , an increasing or decreasing function?

35 Consider the function  $y = \left(\frac{1}{2}\right)^x$ .

- a Determine whether the following functions are equivalent to  $y = \left(\frac{1}{2}\right)^x$ :
  - i  $y = \frac{1}{2^x}$
  - ii  $y = 2^{-x}$
  - iii  $y = -2^x$
  - iv  $y = -2^{-x}$
- b Describe a transformation that would obtain the graph of  $y = \left(\frac{1}{2}\right)^x$  from the graph of  $y = 2^x$ .
- c Graph the functions  $y = 2^x$  and  $y = \left(\frac{1}{2}\right)^x$  on the same coordinate plane.

36 Consider the function  $y = \left(\frac{1}{3}\right)^x$ .

- a Rewrite the function in the form  $y = k^{-x}$ .
- b Describe a transformation that would obtain the graph of  $y = \left(\frac{1}{3}\right)^x$  from the graph of  $y = 3^x$ .
- c Graph the functions  $y = 3^x$  and  $y = \left(\frac{1}{3}\right)^x$  on the same coordinate plane.

37 Consider the exponentials  $y = 3^x$  and  $y = 2^x$



- a Find the equation of the horizontal asymptote of the curve  $y = 3^x$ .
- b Find the equation of the horizontal asymptote of the curve  $y = 2^x$ .
- c Complete the table of values for  $y = 3^x$ :

|     |    |    |    |   |   |   |   |
|-----|----|----|----|---|---|---|---|
| $x$ | -3 | -2 | -1 | 0 | 1 | 2 | 3 |
| $y$ |    |    |    |   |   |   |   |

- d Complete the table of values for  $y = 2^x$ :

|     |    |    |    |   |   |   |   |
|-----|----|----|----|---|---|---|---|
| $x$ | -3 | -2 | -1 | 0 | 1 | 2 | 3 |
| $y$ |    |    |    |   |   |   |   |

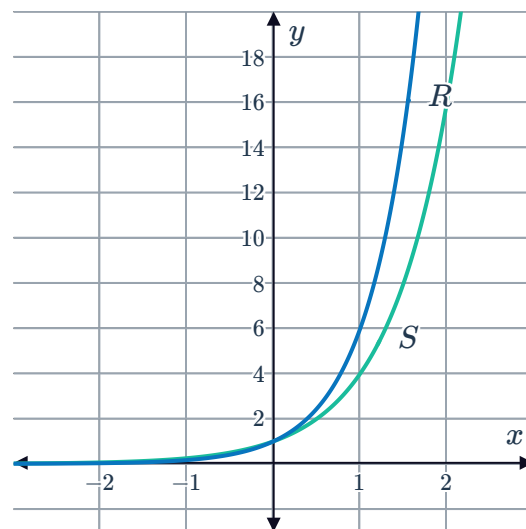
- e Now, sketch the graphs of  $y = 3^x$  and  $y = 2^x$  on the same coordinate plane.
- f State the coordinates of the point at which the two graphs intersect.

38 Consider the functions  $y = 2^x$ ,  $y = 3^x$  and  $y = 5^x$ .

- a Sketch the three functions on the same coordinate plane.
- b State the  $y$ -intercept of each curve.
- c Describe the nature of these functions for increasingly large values of  $x$ .

39 Consider the graphs of the two exponential functions  $R$  and  $S$ :

- a One of the graphs is of  $y = 4^x$  and the other graph is of  $y = 6^x$ . Which is the graph of  $y = 6^x$ ?
- b For  $x < 0$ , is the graph of  $y = 6^x$  above or below the graph of  $y = 4^x$ . Explain your answer.



40 Consider the equation  $y = -10^x$ .



- a Jenny thinks she has found a set of solutions for the equation as shown in the table:

|     |                  |                 |    |     |      |       |
|-----|------------------|-----------------|----|-----|------|-------|
| $x$ | -2               | -1              | 0  | 1   | 2    | 3     |
| $y$ | $-\frac{1}{100}$ | $-\frac{1}{10}$ | -1 | -10 | -100 | -1000 |

She notices that all the  $y$ -values are negative and concludes that for any value of  $x$ ,  $y$  must always be negative. Is she correct? Explain your answer.

- b Sketch the graph of  $y = -10^x$ .
- c Find the values of  $x$  for which  $y = 0$ .